TrustGate IoT

PS-004: Authenticated Gateway-Assisted DNSSEC for IoT Clients

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AIORI Hackathon 3 | Cloud Computing & IoT

Problem Description

• IoT devices use DNS to find the addresses of cloud servers. Forged DNS answers can redirect devices to the wrong destination.

• DNSSEC checks DNS authenticity using digital signatures, but validation adds processing and memory demands.

• We propose one lightweight IoT client connected to one trusted local gateway that performs DNSSEC validation.

• The client authenticates the gateway over an encrypted connection. We will test rejection of bogus DNS answers and fake gateways.

• We will measure latency, client CPU time and memory against a baseline.

What the Device Must Trust

- **DNS:** translates a server name into an IP address. A forged answer can misdirect an application.
- **DNSSEC:** authenticates DNS data using signatures and a configured chain of trust; it does not encrypt queries. [1]
- **DNS over TLS:** encrypts the client-to-gateway channel. Certificate and identity checks are essential to authenticate the gateway. [2, 3]
- **Trust boundary:** the client trusts the gateway's validation result. The client does not independently perform end-to-end DNSSEC validation.
- **Limitations:** a compromised trusted gateway, denial of service and compromised endpoints remain outside this prototype's protection. Application TLS is still needed.

Standards Behind the Design

Reference	Relevant guidance	Design decision
[1] RFC 4035	DNSSEC validation, trust anchors and handling validation failures.	Use an established validating resolver; reject bogus responses.
[2] RFC 7858	DNS over TLS transport and connection reuse.	Use TLS for client queries; measure new and reused connections separately.
[3] RFC 8310	Authentication profiles for encrypted DNS.	Provision gateway identity and trust material; no plaintext fallback.
[4] Unbound documentation	Resolver validation, trust-anchor and TLS configuration.	Configure Unbound on a local gateway and retain validation logs.

Contribution: an IoT-focused implementation and reproducible evaluation of this established model.

TrustGate IoT: Secure Delegation

- **Client:** one lightweight IoT client that sends DNS queries and verifies the gateway identity over TLS.
- **Gateway:** a local system running Unbound. Preconfigure the client with the gateway address, expected identity and trusted certificate authority.
- **Controlled DNS lab:** signed parent zone with a local trust anchor; signed child for valid and tampered answers; unsigned child with authenticated absence of a DS record.
- **Policy:** accept secure answers; identify unsigned answers separately and block them in the strict demo policy. Reject bogus answers and authentication failures.
- **Failure handling:** fail closed on timeout or TLS errors. Use resolver diagnostics to distinguish DNSSEC failure from other server errors.

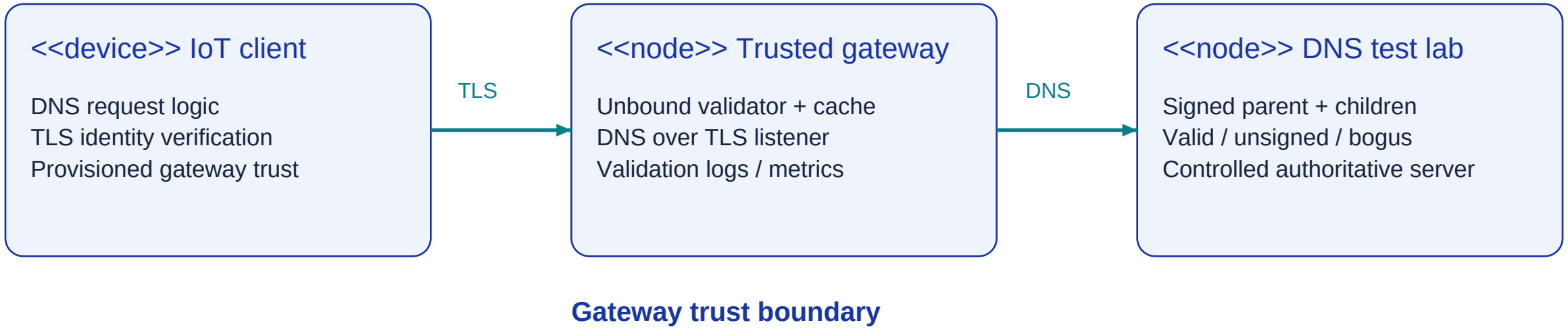
Optimizations and Evaluation Plan

Area	Baseline / before	Proposed approach
Validation work	Client-side validation, if feasible	Gateway validates; measure client resource cost.
Transport	Ordinary DNS to the same gateway	Authenticated TLS; quantify added overhead.
Connection setup	A new TLS connection per query	Reuse the connection to amortize handshake cost.
Repeated lookups	Cold cache	TTL-respecting gateway cache; report warm/cold separately.

Measurement plan: at least 100 lookups per condition; report median and p95 latency, client CPU time and peak memory. Keep device, query set and network constant.

Ordinary DNS measures overhead, not offloading savings. Offloading savings require a comparable client-validation baseline. Measure energy only with suitable hardware.

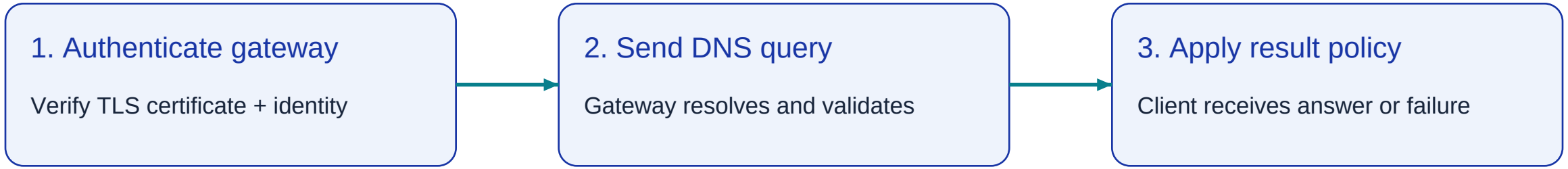
Solution Architecture & Design



The client trusts the authenticated gateway. The gateway validates DNS data against its configured trust anchor. DNS data outside this boundary is not trusted merely because it arrived.

Local lab trust anchors demonstrate a controlled chain of trust; they do not establish public DNS trust. Gateway keys, clock and configuration must be protected.

Query Flow and Expected Outcomes



Test case	Expected result
Secure: valid signed data	Accept validated answer.
Insecure: authenticated unsigned delegation	Identify as insecure; block under strict demo policy.
Bogus: tampered data or invalid signature	Gateway rejects; return failure and record validation reason.
Untrusted gateway certificate / wrong identity	Abort TLS connection before any DNS query is sent.

Client logic: authenticate -> query -> check response -> accept only policy-approved data. Never treat every SERVFAIL as proof of a DNSSEC attack.

Planned Milestones and Methodology

Phase	Planned work	Completion evidence
1. Lab setup	Select client; configure gateway, signed test hierarchy and trust anchor.	Repeatable DNSSEC validation in local lab.
2. Secure integration	Provision gateway identity; implement TLS client and strict policy.	Valid gateway accepted; fake gateway refused.
3. Security tests	Run secure, insecure, bogus and network-failure cases.	Saved logs and expected/actual outcome table.
4. Benchmarking	Run baseline and proposed setup; separate cache and TLS states.	Latency, CPU and memory measurements.
5. Submission	Prepare source, configuration, report, pseudocode and demo.	Reproducible repository and final presentation.

References

[1] R. Arends et al. RFC 4035 (2005). Protocol Modifications for the DNS Security Extensions.

https://www.rfc-editor.org/rfc/rfc4035.html

[2] Z. Hu et al. RFC 7858 (2016). Specification for DNS over Transport Layer Security (TLS).

https://www.rfc-editor.org/rfc/rfc7858.html

[3] S. Dickinson et al. RFC 8310 (2018). Usage Profiles for DNS over TLS and DNS over DTLS.

https://www.rfc-editor.org/rfc/rfc8310.html

[4] NLnet Labs. Unbound documentation. unbound.conf: validator, trust-anchor and TLS configuration.

https://unbound.docs.nlnetlabs.nl/en/latest/manpages/unbound.conf.html

Project brief: AIORI Hackathon 3, PS-004.

Thank You

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Trustworthy DNS resolution for resource-constrained IoT devices.

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